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Dennis Bontempi
Mass General Brigham | Harvard Medical School
Leonard Nuernberg
Mass General Brigham | Harvard Medical School
Deepa Krishnaswamy
Brigham and Women's Hospital
Ahmed Hosny
Mass General Brigham | Harvard Medical School
Keyvan Farahani
NIH NCI
Ron Kikinis
Brigham and Women's Hospital
Andrey Fedorov
Surgical Planning Laboratory, Brigham and Women's Hospital and Harvard Medical School, Harvard University
Hugo JWL Aerts
HAerts@bwh.harvard.edu

Mass General Brigham | Harvard Medical School https://orcid.org/0000-0002-2122-2003

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Transparent and Reproducible AI-based Medical Imaging Pipelines Using the Cloud

Authors | Dennis Bontempi[1,2,3], Leonard Nuernberg[1,2,3], Deepa Krishnaswamy[5], Ahmed Hosny[1,3], Keyvan Farahani[5], Ron Kikinis[5], Andrey Fedorov[5], Hugo JWL Aerts[1,2,3,4]

Affiliations | [1] Artificial Intelligence in Medicine (AIM) Program, Mass General Brigham, Harvard Medical School, Boston, United States of America. [2] Radiology and Nuclear Medicine, CARIM & GROW, Maastricht University, Maastricht, The Netherlands.[3] Department of Radiation Oncology, Brigham and Women's Hospital, Dana-Farber Cancer Institute, Harvard Medical School, Boston, United States of America. [4] Department of Radiology, Brigham and Women's Hospital, Harvard Medical School, Boston, United States of America.[5] National Heart, Lung, and Blood Institute and National Cancer Institute, Bethesda, Maryland

Running title | Cloud Empowers Reproducible AI Medical Imaging

Corresponding author | Hugo Aerts, Ph.D. Director, Artificial Intelligence in Medicine (AIM) Program, Mass General Brigham, Harvard University, Harvard Institutes of Medicine – HIM 343, 77 Avenue Louis Pasteur, Boston, MA 02115. P - 617.525.7156, F - 617.582.6037, Email: haerts@bwh.harvard.edu

Summary | Artificial intelligence (AI) algorithms hold the potential to revolutionize medical imaging. However, a significant portion of the published literature lacks transparency and reproducibility, which hampers sustained progress toward clinical translation. Although several reporting guidelines have been proposed, identifying practical means to address this issue remains challenging. Here, we discuss the potential of cloud-based infrastructures for implementing and sharing transparent and reproducible AI-based medical imaging pipelines. Starting from a published cancer imaging study investigating an AI biomarker for the prognostication of non-small cell lung cancer patients, we demonstrate end-to-end reproducibility of the research from retrieving cloud-hosted data, through data pre-processing, deep learning inference, and post-processing, to the analysis and reporting of the final results. By leveraging cloud-based platforms, using data hosted by the Imaging Data Commons (IDC) combined with code and compute resources, we make every step of the analysis and the resulting data easy to access, reproduce, and share. Our approach demonstrates the potential of cloud resources for implementing, sharing, and using reproducible and transparent AI pipelines in medical imaging, which can accelerate the translation into clinical solutions.

Introduction | Recent advances in artificial intelligence (AI) are helping to address many challenges in different medical areas, such as radiology, radiation oncology, and pathology[1–4]. Several factors contributed to these notable AI breakthroughs, including the progressive increase in computational processing power driven by the development of specialized hardware, the recent advances in deep learning (DL) algorithms, the availability of widely used open-source DL platforms, and the ever-growing amount of publicly available data to train and validate AI models on. Nonetheless, the lack of transparency and reproducibility of published research, as well as processing pipelines, remains one of the biggest challenges faced by the field[5]. Recent literature[6,7] suggest that as little as one-quarter of the AI studies share all the resources to replicate the findings as reported. Although this so-called reproducibility crisis is

not exclusive to AI[8], the sheer complexity and black-box nature of DL algorithms, the large amount of data required to develop and validate models with millions of parameters, and the poor documentation and sharing of AI research[5] substantially worsen the problem.

Transparency and reproducibility are of utmost importance, particularly in the medical field, where errors may result in unpredictable and potentially untoward consequences. While AI can improve the efficiency and effectiveness of patient care,[9] poor implementations can introduce significant biases, exacerbate health disparities[10], and even lead to misdiagnoses and suboptimal treatments[11,12]. Inadequate efforts in making AI research reproducible will make performance claims very difficult to verify[13], ultimately leading to inflated accuracy rates[14] and generalizability problems[15,16], hindering the translation of these systems to the clinic[17]. Furthermore, such a lack of reproducibility often impedes comparing performance across different AI tools.

Several reporting guidelines, checklists, and standards have been proposed in recent years to tackle these issues in the medical imaging field[18–24]. These protocols aim to instruct scientists on the best practices in documenting their research and help them identify aspects of the AI-based analyses that should be reported in publications to ensure reproducibility. Moreover, community-driven efforts led to the creation of other resources, such as a registry of AI models developed for biomedical applications[25] that researchers can use to report minimal attributes of these systems. While these efforts present authors with the rules to follow during the experimental design process and manuscript preparation, they do not provide a practical way to share AI-based pipelines with the community. Even if part of the code is shared, building the pipeline to reproduce the results reported in a study is often “left as an exercise for the reader.” Furthermore, these pipelines often need domain-specific tools, specific software versions, or hardware capability to be reproduced - or even run - as intended without encountering errors, with this information being regularly withheld from the reader. Therefore, without the accurate implementation of all the steps, any community attempt to expand on a study or to identify critical issues with (and failing conditions for) a pipeline

remains practically impossible. Overall, in contrast to other research fields being transformed by technologies - e.g., genomics, where the MIAME guidelines[26] and efforts like the NCI Genomics Data Commons have been instrumental in improving reproducibility by establishing standardized reporting and facilitating data sharing[27,28] - the lack of reproducibility in AI research for medical imaging is significantly more widespread[29].

Cloud-based resources can facilitate a solution to these issues and improve the reproducibility and scalability of AI studies by revolutionizing the way researchers access and use data in their studies. The use of cloud platforms can facilitate access to large-scale cancer imaging data while delivering a more streamlined and consistent computing environment to conduct experiments. Furthermore, by providing public access to data, software, and hardware, cloud-based resources can enable researchers to inspect all the components needed to replicate AI studies (see **Figure 1**), ultimately contributing to the growth and advancement of the field.

In this study, we discuss a cloud-based infrastructure for implementing and sharing end-to-end reproducible and transparent AI pipelines in medical imaging. On the basis of a representative use case, we show end-to-end replication of a published AI study developing and validating an algorithm to enable outcome prediction in patients with non-small cell lung cancer (NSCLC)[30]. By leveraging the National Cancer Institute's (CRDC) Imaging Data Commons (IDC)[31] component for data and the Google Cloud Platform (GCP) component for code and computing, each step within the AI pipeline - from data fetching and pre-processing, deep learning inference, data post-processing, and the analysis of the final results - can be examined and reproduced. Our approach demonstrates the potential of cloud-based resources for implementing, sharing, and using reproducible and transparent AI pipelines, which can accelerate the translation of AI algorithms into the clinic.

Main text | Here, we illustrate a general cloud-based workflow for medical imaging AI using the approach proposed in **Figure 2**. The Imaging Data Commons (IDC) can accelerate

research by centralizing and providing easy access to a wide range of standardized medical imaging data. The cloud-based platform enables validation of medical imaging use cases using public datasets by allowing users to create cohorts of imaging study and image-derived data based on DICOM metadata. Furthermore, since IDC supports data versioning, the specific cohorts remain accessible at any point in time, ensuring long-term preservation and accessibility of imaging data for reproducibility and future research while helping to associate a set of processing results with a specific cohort version.

Cloud-based AI Pipelines can be implemented using computing infrastructures such as Google Colaboratory, among others. These cloud-based services provide users with a Jupyter Notebook environment for writing and running Python code alongside markdown annotation from a web browser - enabling researchers to easily share minimal working examples or full-fledged analyses, ensuring that others can replicate their findings accurately. Notebooks are especially suited to bolster transparency and reproducibility, as they allow for the integration of text, visualizations, and code in a single document, facilitating precise and comprehensive communication of the tools used, the inner workings of the data preparation and processing pipeline, and the research findings. Moreover, their procedural structure makes it easier for new users to comprehend and follow the pipeline workflow and eliminate the need for users to set up their own runtime environment. Finally, their integration with platforms like GitHub streamlines the development and collaboration process.

Cloud-based infrastructure also provides researchers with simplified access to analyses and methods, increasing the reliability and validity of research findings and boosting confidence in reported results. Tools like Google Colaboratory offer descriptive text and interactive visualizations alongside code, creating an ideal data exploration and analysis environment, especially for complex pipelines. The consistent computing environment and access to predetermined resources also ensure transparency in the reporting process concerning computational costs.

To demonstrate the proposed approach, we implemented a case study related to cancer imaging. We selected one of the field's first and most cited studies, presenting an innovative approach showing how convolutional neural networks (CNN) could be used to help improve lung cancer patient stratification[30]. In this investigation, the authors used both clinical and imaging data (standard-of-care computed tomography (CT) scans) from diverse cohorts of non-small cell lung cancer (NSCLC) imaging studies to assess the performance of deep learning for lung cancer prognostication, comparing the latter to conventional prognostic factors and other machine learning methodologies such as radiomics. This comparison was carried out using several metrics and survival analysis tools, such as the area under the receiver operating characteristic (ROC) curve (AUC) and Kaplain-Meier analysis, with the study concluding that deep learning techniques could provide valuable insights into lung cancer prognosis and potentially assist in personalized treatment decisions. The researchers trained a CNN for predicting outcomes by analyzing tumor lesions on CT images and shared the model with the original investigation, together with the AI-derived prognostic scores for the patients analyzed in the study. Although the authors shared an open-source implementation of the algorithm, the trained model weights, the code to run inference using the latter, and a written description of how to prepare data for inference, the data pre-processing component was not included, leaving the users with no alternative but to implement the missing pieces themselves – a step that, as thoroughly investigated, can substantially change the result of computational pipelines when ill-executed[5,32–34]. Besides its impact on the field, another reason we chose this study for our use case was the fact that the AI pipeline was validated using an independent cohort now hosted by the IDC, i.e., the NSCLC-Radiomics dataset[35], allowing us to use the platform to reproduce the AI pipeline validation.

We re-implemented the AI workflow using Google Colaboratory, sharing additional resources researchers can use to replicate and build upon the study and highlighting how the cloud-based infrastructure addresses the previously mentioned issues related to transparency and reproducibility. We provide an overview of the re-implementation in **Figure 3**. We used

136 the IDC platform to define a data cohort matching the subset of the NSCLC-Radiomics dataset
137 used in the original publication. By querying the accompanying metadata curated in IDC tables
138 using SQL, we retrieved the necessary imaging data for the cohort in a transparent and
139 reproducible manner. We make the query to generate the imaging data cohort (and its result)
140 publicly available within the notebooks, together with a description of the procedure used to
141 select the validation cohorts for our study. The resources we share also describe the details
142 and provide an example of how to run the queries to generate the validation cohort. This
143 makes it easier for others to replicate or extend the study by referring to the versioned IDC
144 cohort.

145 Following the textual description provided by the authors, we implemented a pre-
146 processing pipeline for converting the imaging data retrieved from IDC to a format suited for
147 the deep-learning pipeline, relying exclusively on well-adopted open-source tools regarded as
148 robust by the imaging community[36,37]. These software tools are packaged and versioned and
149 can thus be recovered and installed at any point in time. To promote forward compatibility, we
150 also converted the original model weights in the open neural network exchange (ONNX)[38]. We
151 make the code for the model conversion and additional details available as part of our project
152 repository, together with a notebook that incorporates a comprehensive description of every
153 step of the replication process to help users navigate the complexity of the pipeline. Finally,
154 we share a notebook with the code used to generate the results we present in this
155 publication[39].

156 In order to replicate the tumor lesion analysis with the model by Hosny et Al., we used
157 expert segmentations included in the NSCLC-Radiomics collection, also hosted by IDC. As
158 these masks are a newer iteration of the set of labels used for the original publication, we
159 anticipated differences in the results. As reported in **Figure 3b**, the AUC of the original
160 pipeline, computed on the cohort of N=206 subjects, was AUC=0.7, while the replicated
161 pipeline scored AUC=0.68. After testing with various statistical tools, we observed that the
162 difference between the results produced by the pipeline originally published by Hosny et Al.

and those generated by our pipeline are not statistically significant (two-sided Mann-Whitney U test $p>0.05$, $N=206$; DeLong test for correlated (paired) AUC curves $p>0.05$, $N=206$). Therefore, we can conclude that the discrepancy between the two predictive models is negligible. These results also prove the model's robustness to variation in the input segmentation mask, as the original work claims.

Furthermore, we conducted a Kaplan-Meier analysis, as done in the original publication, to assess the stratification power of the AI pipeline. We found that both the original pipeline and the replicated pipeline can successfully stratify higher-risk patients from lower-risk patients ($p<0.001$ and $p=0.023$, for the original and the replicated pipeline, respectively; see **Figure 3b**) when the risk-score threshold shared with the original publication is used to compute the split. In the project repository, we share a notebook for the visualization, exploration, and interpretation of the results of this analysis. The users can use this notebook to replicate the quantitative findings resulting from the comparison of the two AI pipelines and to investigate all the details of our analyses[39]. Finally, the resources we provide can be easily extended to run the same analysis on any other collection hosted on the Imaging Data Commons, provided the dataset contains imaging data suited to the AI pipeline.

Discussion | In this paper, we showcased the potential of cloud-based infrastructures for implementing and sharing transparent and reproducible AI-based medical imaging pipelines. We demonstrated how cloud-based resources offer potential solutions to issues surrounding the reproducibility of AI studies by providing a consistent computing environment, simplifying data exploration and access, and enabling the storage and sharing of code and results. We highlighted how the IDC and the GCP offer easily accessible cloud-based data resources that facilitate the implementation and sharing of fully reproducible AI pipelines in medical imaging. This, paired with cloud-based resources like Google Colaboratory, makes sharing minimal working examples (MWE) of processing pipelines or full-fledged analyses feasible, enabling others to replicate findings correctly while promoting best practices and strategies for data

harmonization. We demonstrated that the proposed cloud-based infrastructure supports the replication of published AI literature.

While cloud-based infrastructure facilitates potential solutions to issues surrounding the reproducibility of AI studies, its adoption also comes with several challenges, the biggest of which is the learning curve for the user unaccustomed to the different interacting blocks and tools. Furthermore, as the value of cloud-based data resources increases with the amount of data hosted by such platforms, any friction in the growth of such resources might constitute a hurdle in the way of their adoption. In the case of platforms such as the IDC, this process may constitute a challenge for a number of users, as additional steps for anonymization and institutional approval may be required to dismiss security and privacy concerns regarding the sharing of protected health information data like medical imaging and the associated clinical data. Nevertheless, the cloud represents a significant advantage for researchers who may not have access to in-house data (or cannot generate data at the same scale as cloud-based data resources) and can help overcome system limitations often encountered in large-scale research studies.

As previously mentioned, one of the benefits of cloud computing is easy and ubiquitous access to a computational environment that requires no experience to be set up and maintained, which researchers can leverage without upfront investments in hardware or infrastructure. While this can be particularly advantageous for small research teams or individuals who may not have access to powerful computing resources, using cloud computing can lead to dependence on third-party providers, which can be a concern for researchers who may need to access resources and data in the long term. Cloud computing providers may change their pricing models or even discontinue services, which could disrupt the research workflow. Additionally, cloud computing may limit the control and flexibility of the user in some regards. For example, researchers may be limited in using obsolete code that requires specific hardware or software - although this problem is not limited to cloud resources but can easily extend to local computational clusters. Furthermore, cloud providers might update these

components without appropriately informing the users, resulting in the disruption, or even failure, of the previously developed pipelines. Cloud computing may not always be the most cost-efficient solution for all use cases, particularly for long-term or high-usage applications.

Finally, while adopting cloud technologies can greatly benefit the medical AI research community, it requires researchers to learn new skills and adapt to a new computing environment. The learning curve can vary depending on a researcher's background and familiarity with cloud computing, and it typically involves gaining an understanding of cloud infrastructure and services and learning how to use cloud-based tools for data storage, processing, and analysis. However, many cloud providers and cloud resources (such as Google Colaboratory, GCP, the CRDC, and IDC) offer comprehensive documentation, training material, and access to a growing community of practitioners to provide guidance and support. With time and practice, the use of cloud technologies can become second nature for researchers, enabling them to take full advantage of the benefits of cloud computing in medical AI research.

In summary, cloud computing offers several advantages for reproducibility in AI research, including democratizing access to large standardized datasets and to free to use, easily accessible, consistent computational environments. There still are essential factors to consider when using cloud computing in AI research, often cost-related - but we believe embracing cloud technology could reduce its costs and ultimately yield the providers of cloud computing resources to develop more customized and cost-effective solutions for the medical image analysis research community. Finally, the long-term cost savings of cloud adoption span beyond direct infrastructure and computing expenses. By enabling greater collaboration and data sharing, cloud technology can accelerate medical AI research and the development of more accurate, effective, and personalized clinical tools. This underscores the compelling case for embracing cloud technology in medical AI research.

DATA AND CODE AVAILABILITY

All of the pipeline components are versioned and can be run and accessed in the cloud.

The version of the NSCLC-Radiomics dataset used for the study can be explored and obtained from the Imaging Data Commons platform (https://portal.imaging.datacommons.cancer.gov).

The codebase and the relative documentation can be accessed at the project repository https://github.com/ImagingDataCommons/idc-radiomics-reproducibility/releases/tag/v1.0.0 and referred to through Zenodo (https://zenodo.org/record/8073712; https://doi.org/10.5281/zenodo.8073712).

The cloud implementation of the pipeline can be accessed and run through the notebooks found at https://github.com/ImagingDataCommons/idc-radiomics-reproducibility/tree/main/notebooks:

- The minimal working example of the cloud pipeline can be run at https://colab.research.google.com/github/ImagingDataCommons/idc-radiomics-reproducibility/blob/main/notebooks/processing_example.ipynb
- The complete model validation cloud pipeline can be run at https://colab.research.google.com/github/ImagingDataCommons/idc-radiomics-reproducibility/blob/main/notebooks/complete_inference.ipynb
- The results analysis, generating all of the plots used in the manuscript figures, can be run at https://colab.research.google.com/github/ImagingDataCommons/idc-radiomics-reproducibility/blob/main/notebooks/results_comparison.ipynb

COMPETING INTERESTS

The authors declare no other competing interests in this work.

AUTHOR CONTRIBUTIONS

Study conceptualization: D.B., H.A., A.F.
Code implementation and result analyses: D.B.
Writing of the manuscript: D.B., L.N., H.A., A.F., D.K.
Critical revision of the manuscript: All authors
Study supervision: H.A., A.F.

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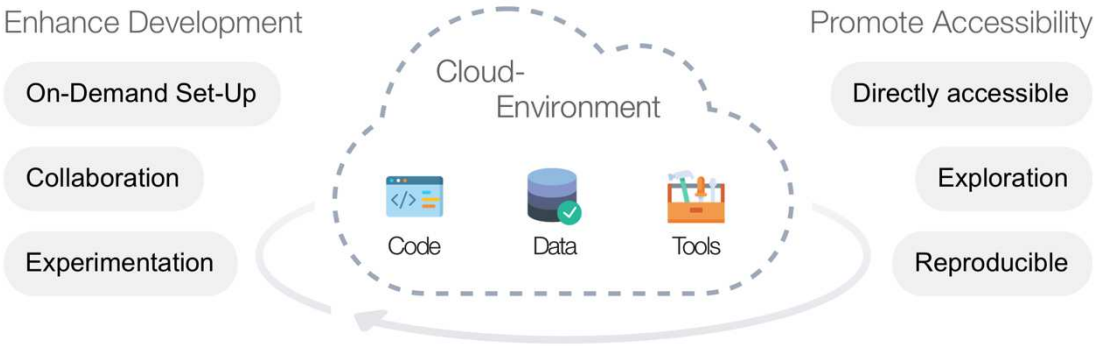

Figure 1 | Potential of cloud-based resources for reproducible AI research. Cloud-based infrastructures offer solutions to challenges related to reproducibility by providing access to large standardized datasets, resources for data exploration, and standardized environments for code and hardware. Cloud-based resources can simplify data exploration by enabling researchers to explore data easily and overcome system limitations often encountered in local research studies. Cloud-based infrastructures can also facilitate storing and sharing code and results, promoting researchers to reproduce and verify results. Finally, cloud-based computational infrastructures offer a uniform environment, equipped with pre-built libraries and tools, for researchers to conduct their experiments more transparently and reproducibly.

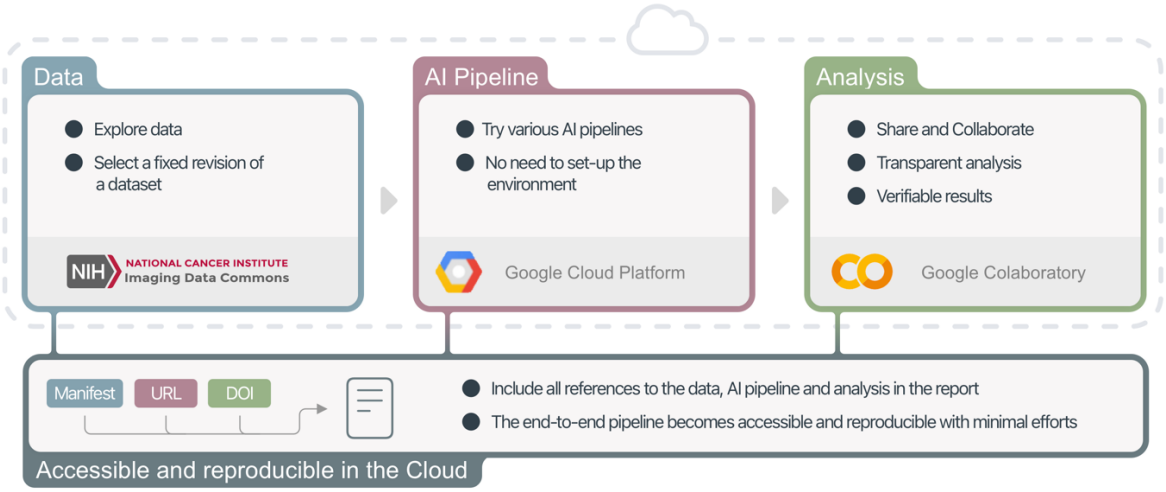

Figure 2 | Cloud-based workflow for medical imaging AI. Our proposed approach combines open-source components, commercially available tools, and infrastructure being established by the US National Cancer Institute (NCI) to implement reproducible image analysis workflows and simplify their external/independent evaluation. We propose using the NCI Imaging Data Commons (IDC) as the persistent location of public imaging data, ensuring that data is available for public use, harmonized, and can be accessed in a standardized fashion. To facilitate computational workflows, we propose using components from the Google Cloud Platform to provision preconfigured GPU-enabled resources for the execution of AI pipelines. To promote transparency and reproducibility, we propose the use of Google Colaboratory to evaluate the results of the AI pipeline; by providing a ready-to-use, cloud-based Jupyter Notebook environment, this eliminates the need for local software installation and configuration, allowing for easier verification and sharing of the results. The combination of these different tools ensures an easily accessible end-to-end pipeline that can be inspected and easily referenced in all of its components.

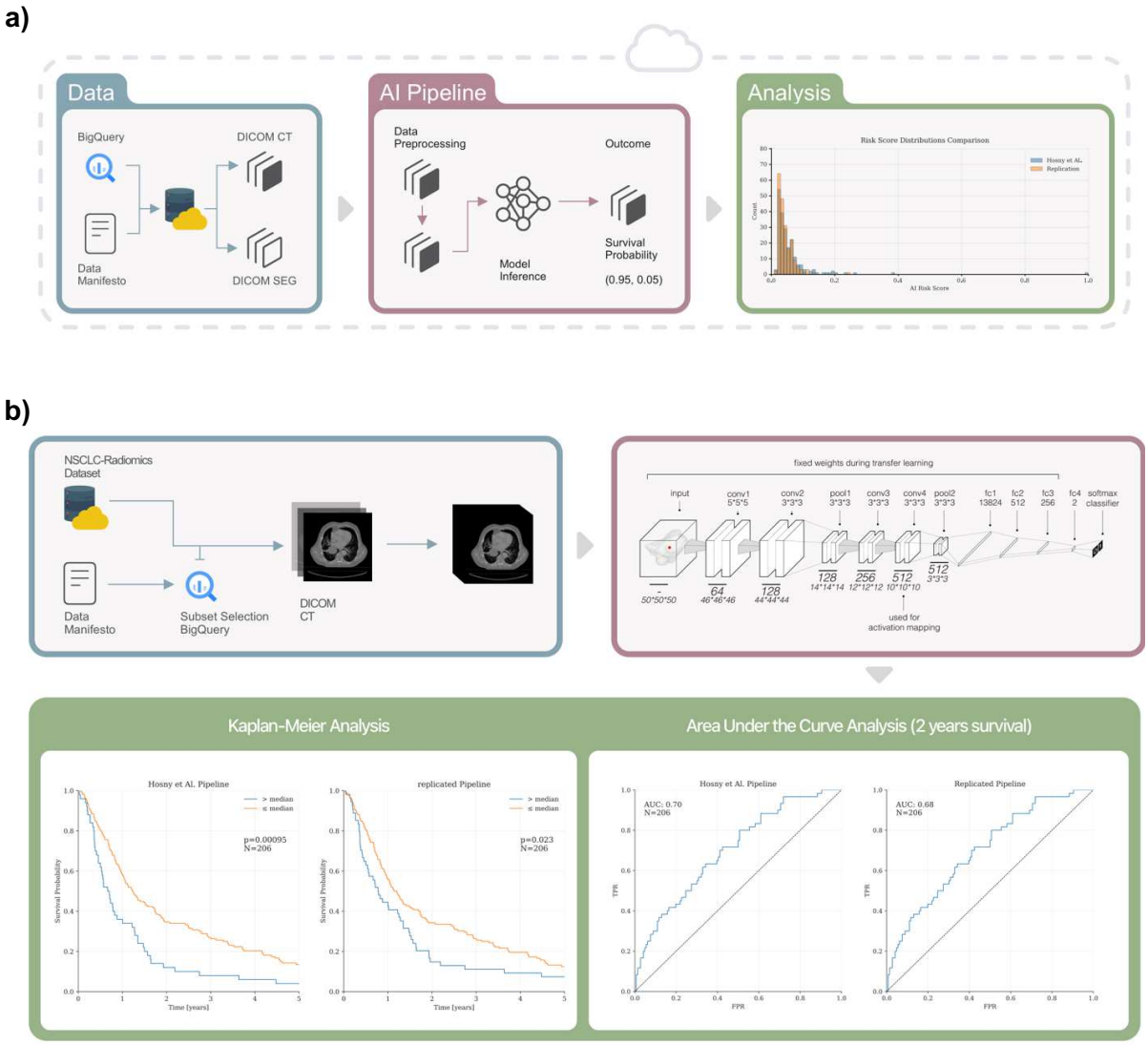

Figure 3 | Cloud-based re-implementation of a published study. ***a)*** We demonstrate our proposed approach by re-implementing a cancer imaging analysis pipeline from Hosny et al. (PLOS Medicine, 2018) using cloud-based infrastructure. We obtained the data from the Imaging Data Commons and used Google Colab for computation, showcasing the benefits of our proposed cloud-based infrastructure for improving transparency and reproducibility. By leveraging the cloud, we are able to share every step of the pipeline with the community, from data fetching to the reporting of results, enabling researchers to easily verify, replicate and build upon the study. To provide a practical example of how the proposed approach can be useful for bolstering transparency and reproducibility, **b)** we compared our cloud-based pipeline with the original results shared by Hosny et Al. on a subset of the NSCLC-Radiomics dataset, which is made easily accessible thanks to the functionalities of the IDC platform. We found no statistically significant difference in the area under the curve (AUC), and we observed that the split based on Hosny et Al. AI risk score is still significant and able to stratify patients at higher risk. All the replication steps, including the code to generate this figure, are shared publicly at the project repository and can be accessed and run for free on the cloud.